Xinyi Huang

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EDUCATION

The University of Hong Kong Master's in Electrical and Electronic Engineering 2025.09-2026.11

- GPA: 3.36/4.00; Electrical and Electronic Engineering Scholarship
- Course: Integrated Circuits, Chip & Memory Design, AI and Edge Processors, Data Compression

Beijing Jiaotong University B.Eng. in Communication Engineering

- GPA: 3.26/4.0 (Top 17%); Third-Class Academic Excellence Scholarship 2021.09-2025.06
- Course: Digital Circuits, Analog Circuits, Digital Signal Processing, Microprocessor Principles & Interfacing

INTERNSHIP EXPERIENCE

Bitmain 2026.06-Present

Objective: Contributing to optimized FP64 arithmetic and on-chip NoC/DDR data-path architecture for a RandomX virtual-machine workload.

FP64 Division / Square-Root Pipeline Design and Frequency Optimization

- Designed a unified 5-cycle, fixed-latency shared pipeline for FP64 DIV and SQRT using Goldschmidt division, reciprocal-square-root Newton iteration, 57-bit Q2.55 fixed-point arithmetic, and LUT-based initial approximations; supports one new instruction per cycle.
- Removed redundant clock-gating and operand-isolation logic based on DC power reports to reduce power consumption.
- Formally verified key functional and timing properties over bounded numeric ranges using **VC Formal**.
- Achieved timing closure at 2.4 GHz and 2.6 GHz under 5 nm, 0.75 V, 85 C conditions, with no setup violations; results are bit-accurate against the reference model and power increased only 20% versus the prior iterative operator.

NoC Architecture and Cycle-Accurate Performance Modeling

- Evaluated candidate topologies and DDR placements; defined a 2D-mesh NoC with deterministic request/response routing while reusing proven architectural components to reduce verification risk.
- Defined segmented responses, bounded outstanding transactions, FIFO back-pressure, and response-reassembly mechanisms to improve response-path bandwidth utilization.
- Built a parameterized cycle-accurate model covering link pipeline depth, FIFO depth, DDR service latency, and uniform/hotspot/burst traffic to quantify end-to-end latency, hot links, and buffer pressure.

PROJECT EXPERIENCE

Transformer Attention Accelerator SoC Subsystem Design 2025.10-2026.01

Objective: Build a software-programmable dual-clock Attention SoC accelerator subsystem for Transformer inference, covering CPU configuration, DMA transfers, clock-domain crossing, result write-back, and interrupt notification.

Key Contributions:

- Integrated a 500 MHz bus domain with a 1.25 GHz compute domain: AHB-Lite for register configuration and interrupts, 128-bit AXI4-MM DMA for Q/K/V and result transfers, and asynchronous FIFOs for clock-domain crossing.
- Reused an 8 x 8 FP16 systolic array for QK^T and PV; added 8-way parallel Softmax plus score/probability queues so matrix compute, Softmax, and accumulation advance concurrently.
- Reduced dual-clock CDC Error/Fatal findings from 65 to 0 using VC Static.
- Established Python reference-model, VCS regression, SVA, and functional-coverage verification flows.

Results:

- Reduced compute cycles from 342,228 to 79,489 (-76.8%) and end-to-end task cycles from 350,429 to 105,112 (-70.0%) for N=64, d=32 attention.
- Achieved 98.48% functional coverage, 0 SVA failures, and 0 numerical mismatches across four final regressions.
- Completed pre-layout STA for four critical modules using Yosys, ASAP7 RVT TT libraries, and OpenSTA; WNS/TNS = 0 at 1.25 GHz with 50 ps clock uncertainty

SKILLS

HDL/Programming: Verilog, SystemVerilog, C++, Python

EDA: VCS, Verdi, DC, VC Static, Vivado, ModelSim, Cocotb

English: IELTS 6.5